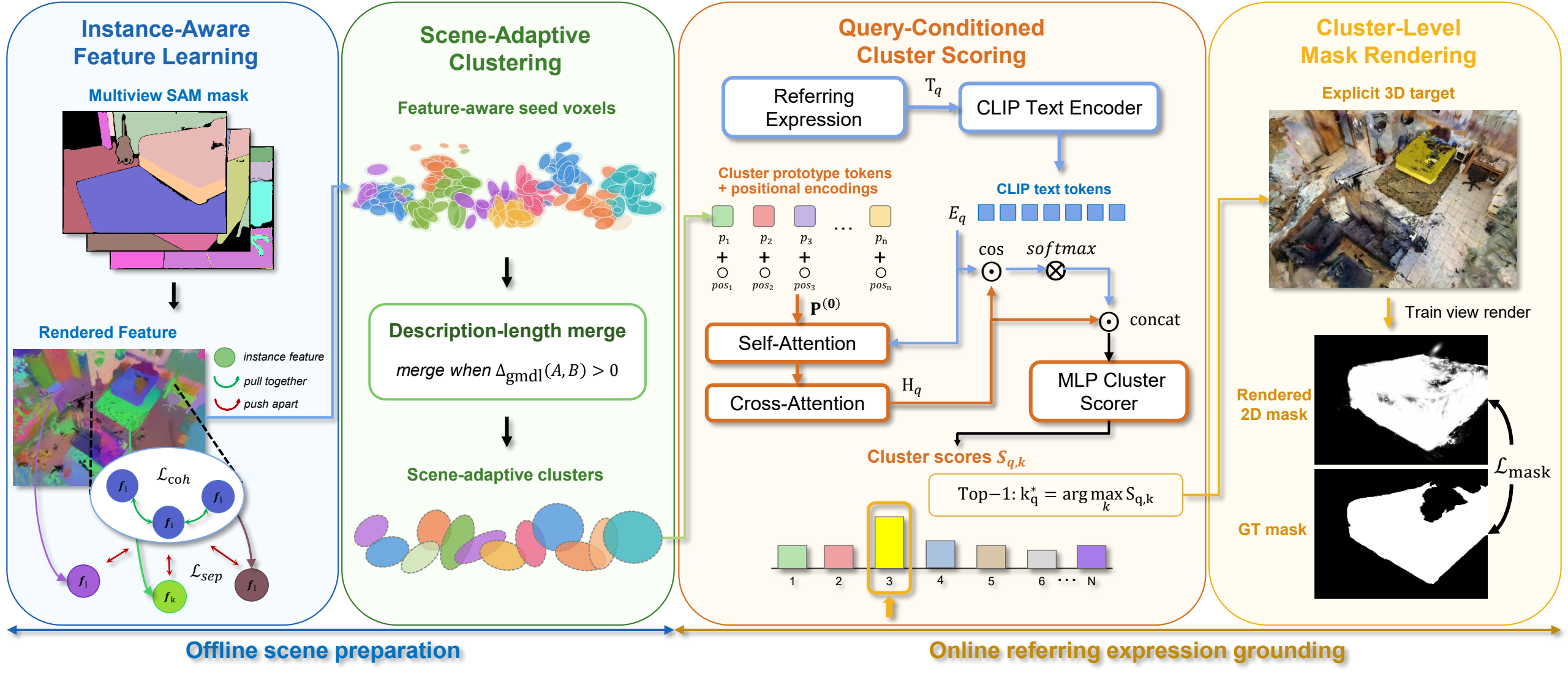

Instance-Aware Feature Learning
Multiview SAM mask
Rendered Feature
instance feature
pull together
push apart
$\mathcal{L}_{coh}$
$\mathcal{L}_{sep}$
Scene-Adaptive Clustering
Feature-aware seed voxels
Description-length merge
merge when $\Delta_{\text{gmdl}}(A, B) > 0$
Scene-adaptive clusters
Query-Conditioned Cluster Scoring
Referring Expression
T_q
CLIP Text Encoder
Cluster prototype tokens + positional encodings
CLIP text tokens
E_q
cos
$softmax$
concat
$\mathbf{P}^{(0)}$
Self-Attention
Cross-Attention
H_q
MLP Cluster Scorer
Cluster scores $S_{q,k}$
Top−1: $k_q^* = \arg\max_k S_{q,k}$
1 2 3 4 5 6 ··· N
Cluster-Level Mask Rendering
Explicit 3D target
Train view render
Rendered 2D mask
GT mask
$\mathcal{L}_{\text{mask}}$
Offline scene preparation
Online referring expression grounding